

Spin Orbit-Torque (SOT) Readers, Amplifiers, and Interconnect Design Configurations

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As magnetic-recording areal densities approach 4 Tb/in², conventional tunneling-magnetoresistance (TMR) read sensors face increasing SNR limitations from reduced track width, rising sensor impedance, and growing noise contributions. Spin-orbit-torque inverse-spin-Hall-effect (SOT-iSHE) readers have been proposed as candidates for extending read-sensor scaling beyond TMR limits. This paper examines low-noise front-end architectures for four-terminal SOT-iSHE readers, emphasizing the interaction between reader, bias network, interconnect, and low-noise amplifier (LNA). Amplifier configurations, biasing approaches, impedance effects, and interconnect options are evaluated for their impact on total readback noise figure (NF). The objective is to identify design tradeoffs that preserve or enhance the intrinsic NF advantage of SOT-iSHE readers for future high-areal-density magnetic recording.

Index Terms— spin-orbit torque, inverse spin Hall effect, low-noise amplifier, magnetic recording, noise figure, preamplifier, read sensor, signal-to-noise ratio.

I. INTRODUCTION

As magnetic-recording areal densities approach 4 Tb/in², conventional TMR read sensors face severe scaling challenges. Track widths below ~20 nm reduce readback signal, while increasing sensor resistance and noise from Johnson, shot, magnetic thermal, interconnect, and preamplifier sources degrade achievable SNR. Consequently, the read sensor, interconnect, LNA, and bias network must be treated as a coupled signal path.

SOT-iSHE readers have been proposed as promising candidates for extending magnetic-recording technology beyond conventional TMR scaling, supported by large inverse spin Hall response and measurable output-voltage levels at scaled dimensions [1].

This work examines low-noise front-end architectures for four-terminal SOT-iSHE readers, evaluating amplifier topology, biasing, impedance, and interconnect effects on total readback NF. The four-terminal structure allows separation of bias and sense paths, enabling techniques that preserve the intrinsic SNR advantage of SOT-iSHE readers. We summarize present status, provide early SNR results, and identify key tradeoffs for SOT-reader integration.

II. SOT-ISHE TRANSDUCER'S SIGNAL AND SNR SCALABILITY

A simplified schematic of a 4-terminal SOT-iSHE reader is shown in Fig. 1(a). In this geometry, a spin current generated by the ferromagnetic free layer (FL) is injected into an adjacent non-magnetic channel (here labeled SHL) where iSHE converts it into a longitudinal charge voltage. Spin-to-charge (S-C) conversions can occur in the bulk, as in heavy metals such as Ta, Pt, or W, and/or at interfaces through interfacial S-C conversions. In the framework of 1D spin diffusion models, one expects the iSHE-generated voltage V_{SHE} to scale as:

$$V_{SHE} \propto TW \theta_{SH} \rho_{SHL} \beta j_c. \quad (1)$$

Here, TW is the width of the FL, θ_{SH} is the spin Hall angle, ρ_{SHL} the resistivity of the SHL layer, β the *effective* spin polarization efficiency of the FL, and j_c the injected perpendicular charge current density. Crucially, V_{SHE} scales linearly with TW as opposed to quadratically (barrier area) in conventional TMR sensors. Additionally, V_{SHE} depends both on the conversion efficiency of spin to charge in the SHL layer (θ_{SH}) and the resistivity of the channel (ρ_{SHL}). Signal increases through higher spin Hall angles come for free as opposed to increases through ρ_{SHL} , that also impact noise levels on the voltage detection path.

In a 4-terminal device, one needs to carefully separate noise sources between the current injection and voltage detection paths. On the current injection path, besides the thermal magnetic noise of the FL (ultimately proportional to V_{SHE}), one needs to consider the preamplifier bias current noise, which will be reflected in a voltage noise through the SHE resistance $R_{SHE} = V_{SHE}/I_c$. On the voltage detection path (considered under no DCbias) noise sources include Johnson noise and the internal preamplifier noise. Both scale with the resistance of the channel (R_{SHL}).

Fig. 2(b) shows typical resistance values considering an all-metallic CoFe-based FL stack. R_{SHL} is an order of magnitude smaller than R_{SHE} , resulting in negligible current bias noise contributions to total noise. On the current injection path noise is dominated by thermal magnetic noise. As R_{SHL} is independent of TW , preamplifier noise does not increase with narrow FL TW s, contrary to conventional TMR-based readers. The proposed low-noise preamplifier architecture, as compared to existing preamplifier designs, allows for a substantial ~5dB SNR gains (see Fig.1(c)).

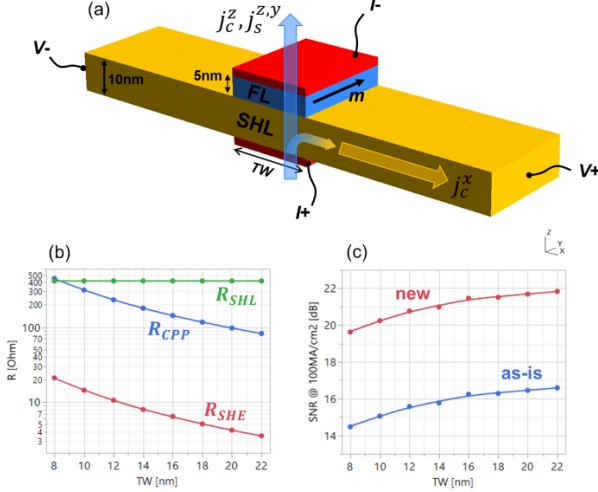


Fig. 1. (a) 4-terminal SOT-iSHE reader schematic (top and bottom shields not shown). (b) Typical resistance values for a purely metallic square-cross-section pillar ($\rho_{FL} = 20\mu\Omega \cdot \text{cm}$; $\rho_{SHL} = 500\mu\Omega \cdot \text{cm}$). (c) Achievable pk-pk SNR (per 100MA/cm² of injected current) vs. TW , comparing as-is and new low-noise preamplifier designs. Using $\theta_{SH} = 1$ for reference.

III. LOW-NOISE PREAMPLIFIER ARCHITECTURE

The four-terminal SOT-iSHE geometry enables preamplifier architectures unavailable in two-terminal TMR readers. The transverse bias flows through separate leads, eliminating dc-bias from the differential signal path and removing bias-current noise coupling from the output.

Fig. 2 shows the proposed SOT bias amplifier with voltage-controlled biasing. A CMOS voltage-sense topology with source followers decouples input impedance from noise optimization [3,4]. Unlike common-base current-sensing, where reducing noise lowers input impedance and worsens T-line mismatch, this topology provides independent impedance control via gain and feedback resistance [4,5].

The interconnect is modeled using transmission (ABCD) chain matrices capturing frequency-dependent T-line impedance transformation. At quarter-wave resonance, impedance inverts ($Z_{in} = Z_0^2/R_s$), producing noise peaking and sensitivity nulls. The SOT having an independent bias path greatly reduces the bias noise coupling into the signal path as opposed to TMR sensors.

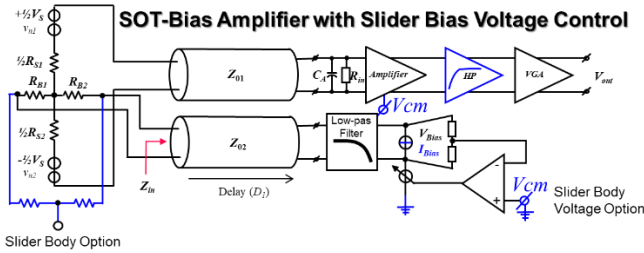


Fig. 2. Proposed SOT-iSHE preamplifier with voltage-controlled bias. The 4-terminal configuration separates the bias path from the differential signal amplification path.

IV. NOISE FIGURE ANALYSIS AND RESULTS

We compare the integrated NF of the proposed SOT-DC architecture against a baseline BiCMOS current-sense preamplifier for two-terminal TMR readers [4]. NF is evaluated using linear two-port noise theory with input-referred voltage and current noise extracted from circuit simulations. T-line effects are incorporated through the ABCD matrix formulation, referring total electronic noise back to the sensor source impedance.

Fig. 3 shows the integrated NF difference between SOT-DC and a typical current-sense configuration. The SOT architecture achieves ~ 1.1 dB improvement across the impedance range from eliminating bias-current noise. An additional ~ 0.5 dB is projected from amplifier re-optimization without bias circuitry constraints.

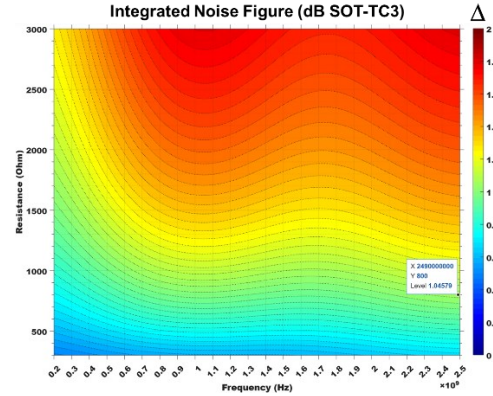


Fig. 3. Integrated NF improvement of SOT-dc preamplifier relative to TC3 baseline versus sensor resistance.

Further gains are anticipated from ac-bias operation using a >2 GHz carrier to eliminate electronic $1/f$ noise; this will be detailed in future work. We will have initial measurements of signal and noise from recent hardware.

V. CONCLUSION

This paper presented a noise-figure analysis of preamplifier architectures for four-terminal SOT-iSHE readers. The four-terminal geometry separates bias and signal paths, eliminating bias-current noise coupling. The proposed voltage-sense SOT-dc architecture demonstrates ~ 1.1 dB integrated NF improvement over an optimized baseline, with the lower SOT channel impedance reducing T-line mismatch. Future work targets additional gain through ac-bias architectures.

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